

FORMATIVE ASSIGNMENT 1

Comparative Analysis of Sequential Models for Mobile Network Traffic Forecasting

Overview

Forecasting sequential data is an important problem in many real-world domains. In this assignment, you will conduct a research-informed empirical investigation into mobile network traffic forecasting.

You will work with telecommunications activity data collected across Milan over approximately two months. The city is divided into 10,000 geographical areas, with observations recorded at 10-minute intervals. The dataset is large enough that efficient data handling and memory management are important parts of the investigation. Download the dataset from [HERE](#)

Your study should address the following research question: “How do different sequential models compare for one-step-ahead mobile network traffic forecasting, and how does their performance vary across geographical areas with different traffic characteristics?”

The objective is not simply to train several models and identify the one with the lowest prediction error. You are expected to investigate the problem, review relevant research, make and justify methodological decisions, conduct experiments, and draw conclusions supported by evidence.

1. Data Handling and Memory Management

Develop an appropriate strategy for loading and processing the dataset within the computational resources available to you.

Describe your approach and the decisions made to manage memory efficiently. Include evidence showing memory usage before and after your optimisation, together with a brief discussion of the improvement achieved and any important limitations or trade-offs.

The specific techniques used are left to you, but you should be able to explain how and why your approach improves the efficiency of the implementation. Where appropriate, support your approach with relevant sources.

2. Exploratory Analysis

Begin by investigating the characteristics of the Internet traffic data.

1. Include a figure showing the distribution of total Internet traffic across the geographical areas during the two-month observation period, accompanied by a discussion of the main characteristics of the distribution.
2. Identify the three geographical areas with the highest total Internet traffic during the full observation period. Include figures showing the Internet traffic time series during the first two weeks for five geographical areas:
 - a. the three areas with the highest total traffic,
 - b. Square ID 4159, and
 - c. Square ID 4556.

Accompany these figures with a discussion of similarities and differences in their temporal dynamics, including notable patterns or unusual behaviour and possible explanations.

Finally, conduct two additional analyses of your choice using the area with the highest total traffic. These analyses should improve your understanding of its temporal characteristics and be relevant to the forecasting problem. Possible areas of investigation include temporal dependencies, periodic behaviour, autocorrelation, stationarity, seasonality, or anomalies.

For each analysis, present the relevant output and explain what it reveals about the traffic behaviour and how the findings may inform your forecasting approach. Focus on analyses that provide useful evidence rather than producing a large number of visualisations.

3. Related Work and Model Selection

The purpose of this section is to identify and justify the three models that you will implement, evaluate, and compare in Section 4. You are not required to implement the models at this stage. Conduct a focused review of relevant research on mobile or cellular network traffic forecasting and/or closely related sequential forecasting problems. Use this review, together with your exploratory analysis, to inform your methodological decisions.

Select three models appropriate based on your related work review for the forecasting problem. The models should be sufficiently different to support a meaningful comparison; minor variations of the same architecture will not be considered different models.

For each model, explain why it was selected. Your justification should consider the characteristics identified in your exploratory analysis, relevant findings from previous research, and your understanding/criticism of the model's strengths and limitations.

For example, if a model is selected because it is expected to capture long-term dependencies, seasonality, nonlinear behaviour, or another characteristic relevant to the data, explain and support this decision with appropriate evidence. Existing research should inform your decisions, but your objective is not to reproduce an existing study. The three models identified in this section are the same models that must be implemented and evaluated in Section 4.

4. Forecasting Experiments

In the third task, you are asked to implement the three models selected in Section 3 for one-step-ahead forecasting of Internet traffic activity for one-step-ahead prediction of network traffic in a single geographical area. Formally, let us denote as $x_a(t)$ the traffic observed at area a during time interval t . At each time t , the algorithm receives as input a history x_t , i.e., a vector of traffic values in past time intervals, up to t included.

The algorithm shall then produce as output an estimate $\bar{x}_a(t + 1)$ of future traffic at $t + 1$ in area a . You shall run the algorithm to forecast traffic in the geographical area with the highest total identified at the second item of Task 2, during the week from December 16 to 22. The following results should be produced as a minimum:

- I. A self-contained description of all proposed models, where each model is clearly described in terms of its structure, input representation, preprocessing steps, and training procedure
- II. Plots reporting the superposed time series of (i) the original traffic and (ii) the predicted traffic in the week from December 16 to 22, for each model and for each of the three geographical areas (a total of 9 plots);

- III. Three tables reporting the performance of all selected models for each of the three geographical areas, where each table presents the Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Squared Error (RMSE) for all models
- IV. Exact statistics on the training and execution time of each model, with details on the process used to compute such statistics and on the hardware on which they were recorded; The reported training and execution times should be representative of the model and may be obtained from one selected area or as an average across the three areas, provided the method used is clearly stated.
- V. Explanation text where you provide personal considerations on the design, performance, and possible margins for improvement of all models.
- VI. You must clearly describe the input representation used for each model, including the sequence length, preprocessing steps, and normalization techniques.
- VII. You must provide a comparative analysis of all models, discussing differences in predictive performance, training time, and suitability for the dataset. The identification of the best-performing model must be supported by both quantitative results and qualitative reasoning based on the observed data characteristics from Task 2

Submissions that describe only a single model or do not provide a complete comparison across all implemented models will be considered incomplete. You can provide additional results that complement those above, only if they are strictly needed to highlight some important properties of the models.

Report MAE and RMSE, and other suitable evaluation metrics of your choice. Include a table comparing the three models across the area with the highest traffic and appropriate figures comparing actual and predicted traffic during the evaluation period.

Your discussion should go beyond identifying the model with the lowest error. Consider whether model performance changes across areas and relate your findings, where appropriate, to the traffic characteristics identified during your analysis and to relevant previous research.

Finally, present at least one example of a period or traffic pattern for which one or more models perform poorly, and discuss possible reasons for the observed behaviour.

References

- [1] Gianni Barlacchi, Marco De Nadai, Roberto Larcher, Antonio Casella, Cristiana Chitic, Giovanni Torrisi, Fabrizio Antonelli, Alessandro Vespignani, Alex Pentland, and Bruno Lepri, "A multi-source dataset of urban life in the city of Milan and the Province of Trentino." *Sci Data* 2, 150055 (2015). <https://doi.org/10.1038/sdata.2015.55>
- [2] <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/EGZHFV>
- [3] <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/QJWLFU>